Feature Description 214/221 04-LZA 701 6014/1 Uen BS

Uplink-Triggered Inter-Frequency Mobility

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1 Uplink-Triggered Inter-Frequency Mobility Overview

> This feature enhances mobility by introducing the measurement of uplink channel quality.

Feature Name	Uplink-Triggered Inter-Frequency Mobility
Feature Identity	FAJ 121 1797
Value Package Name	LTE Base Package
Value Package Identity	FAJ 801 0400
Node Type	Baseband Radio Node
Access Type	LTE
Licensing	License-controlled feature. One license is required for each node.

Summary

The Uplink-Triggered Inter-Frequency Mobility feature enhances mobility by introducing uplink channel quality. It is measured in the eNodeB as SINR. It is an addition to the quantities measured on the downlink channel. This improves the Mobility Control at Poor Coverage feature and the functionality of the following features to increase mobility handling flexibility at poor uplink channel coverage:

- Coverage-Triggered Inter-Frequency Handover
- Coverage-Triggered TD-SCDMA Handover
- Coverage-Triggered WCDMA Handover

The feature also introduces an uplink search zone along the edge of the cell, where UE measurements search for alternative frequencies and cells. If a frequency or cell is detected, a mobility action is initiated.

The feature enables UEs with poor uplink coverage to search for inter-frequency target frequencies. The UE can move to another frequency with inter-frequency handover, or to another RAT with IRAT handover.

In case of inter-frequency handover to E-UTRAN cells, the uplink quality of the target cell is evaluated to avoid handover to a cell with worse or insufficient uplink quality. This evaluation is also performed for downlink-triggered handovers.

The feature moves UEs in critical uplink coverage to other E-UTRAN or IRAT cells with handover or release and redirection.

When this feature is activated together with the Automated Neighbor Relations feature, inter-frequency neighbors can be added in the uplink search zone based on mobility triggers instead of triggering on the dedicated ANR measurements that are used in the rest of the cell.

The feature also enables Automated Neighbor Relations to add inter-frequency neighbors based on mobility triggers in the uplink search zone, instead of triggering on the dedicated ANR measurements used in the rest of the cell. This provides the following benefits:

- ANR can define neighbor cells in such a way that the best cell is identified and used.
- The ANR-initiated measurements are removed in the uplink search zone and replaced with measurements intended for mobility. This means that only the cells used for mobility purposes are added. This reduces the neighbor list to only include cells that are used.

Additional Information

For information on the preferred state and parameter settings of this feature, see RAN Parameter Recommendations Lists.

More information about this feature, and related topics, can be found in the following documentation:

- *3GPP TS 36.300 Overall description; Stage 2*
- *3GPP TS 36.331 Radio Resource Control (RRC); Protocol Specification*

2 Dependencies of Uplink-Triggered Inter-Frequency Mobility

> This feature requires the Mobility Control at Poor Coverage feature to operate.

Features

Table 1 Feature Dependencies

Feature	Relationship	Description
Mobility Control at Poor Coverage (FAJ 121 3013)	Prerequisite	The Mobility Control at Poor Coverage feature must be active for the Uplink-Triggered Inter-Frequency Mobility feature to be operational.
Automated Mobility Optimization (FAJ 121 3035)	Related	The Automated Mobility Optimization feature enables the automatic optimization of mobility parameters. The Uplink-Triggered Inter-Frequency Mobility feature can decrease the number of handovers detected by this feature. This is applicable to too late and too early handovers.
Automated Neighbor Relations (FAJ 121 0497)	Related	The Automated Neighbor Relations feature enables the automatic build and maintenance of neighbor cell list for mobility.
Basic Intelligent Connectivity (FAJ 121 4843)	Related	If both the Uplink-Triggered Inter-Frequency Mobility and the Basic Intelligent Connectivity features are activated, the Uplink-Triggered Inter-Frequency Mobility feature can perform the handover of the UE before Basic Intelligent Connectivity triggers mobility action.
Coverage-Triggered CDMA-eHRPD Session Continuity (FAJ 121 0491)	Related	The Coverage-Triggered CDMA-eHRPD Session Continuity feature provides the possibility to initiate release with redirect to a CDMA-eHRPD cell. With the Uplink-Triggered Inter-Frequency Mobility feature enabled, this can occur when critical uplink coverage is reported for the UE in the node.
Coverage-Triggered GERAN Session Continuity (FAJ 121 0495)	Related	The Coverage-Triggered GERAN Session Continuity feature provides the possibility to initiate release with redirect to a GERAN cell. With the Uplink-Triggered Inter-Frequency Mobility feature enabled, this can occur when critical uplink coverage is reported for the UE in the node.
Coverage-Triggered Inter-Frequency Handover (FAJ 121 0877)	Related	The Coverage-Triggered Inter-Frequency Handover feature provides the possibility to initiate handover to a cell on another LTE frequency. With the Uplink-Triggered Inter-Frequency Mobility feature enabled, this can occur when the UE reports poor uplink coverage in the current cell, or when critical uplink coverage is reported for the UE in the node.
Coverage-Triggered Inter-Frequency Session Continuity (FAJ 121 0797)	Related	The Coverage-Triggered Inter-Frequency Session Continuity feature provides the possibility to initiate release with redirect to a cell on an alternative LTE frequency. With the Uplink-Triggered Inter-Frequency Mobility feature enabled, this can occur when critical uplink coverage is reported for the UE in the node.
Coverage-Triggered WCDMA IRAT Handover (FAJ 121 0897)	Related	The Coverage-Triggered WCDMA IRAT Handover feature provides the possibility to initiate handover to a cell on a WCDMA frequency. With the Uplink-Triggered Inter-Frequency Mobility feature enabled, this can occur when the UE reports poor uplink coverage in the current cell, or when critical uplink coverage

Feature	Relationship	Description
		is reported for the UE in the node.
Coverage-Triggered WCDMA Session Continuity (FAJ 121 0493)	Related	The Coverage-Triggered WCDMA Session Continuity feature provides the possibility to initiate release with redirect to a WCDMA cell. With the Uplink-Triggered Inter-Frequency Mobility feature enabled, this can occur when critical uplink coverage is reported for the UE in the node.
Device-Based Frequency Filtering (FAJ 121 5154)	Related	The Device-Based Frequency Filtering feature is used to configure a list of unsupported frequencies for specific UEs. These frequencies are excluded from UE configuration messages sent by the eNodeB. Unsupported frequencies are configured using the `UnsupportedFreqProfile.unsupportedFreqList` attribute.
High Power UE (FAJ 121 4762)	Related	When Uplink-Triggered Inter-Frequency Mobility is used with the High Power UE feature, the maximum allowed output power of the UE in the target and source cells is taken into account in controlling mobility. This allows high power UEs to be served longer on cells where higher output power is available. Also, this improves handover to target cells where high power UE capability is allowed, and so improves uplink quality and coverage. This is achieved by taking into account the maximum allowed output power of the UE for the time normalized SINR calculation for DMRS, based on the actual Carrier Aggregation configuration and the UE power class on the PCell.
Intra-LTE Inter-Mode Handover (FAJ 121 3042)	Related	Handovers can be initiated between LTE FDD and LTE TDD cells if the Intra-LTE Inter-Mode Handover feature is activated with the Uplink-Triggered Inter-Frequency Mobility feature. With the Uplink-Triggered Inter-Frequency Mobility feature enabled, this can occur when the UE reports poor uplink coverage in the current cell, or when critical uplink coverage is reported for the UE in the node.
LTE Broadcast (FAJ 121 3021)	Related	Point-to-multipoint service based on eMBMS. The Uplink-Triggered Inter-Frequency Mobility does not consider whether a UE sends MBMS Interest Indication. This is to ensure that legacy behavior is followed when downlink link quality has higher priority than the ability to receive MBMS. The Uplink-Triggered Inter-Frequency Mobility feature might move the UE to another frequency with better uplink coverage, even if there is no MBMS in the cell. This occurs if the following conditions are met: – Uplink coverage in the cell is poor. – The UE is in connected mode. – The UE receives MBMS. Note: This occurs even if downlink coverage in the cell is good. UEs can combine downlink only eMBMS transmissions from multiple cells. The UE can be moved even if the user does not experience bad video quality. Note: The uplink quality thresholds are defined in the

Feature	Relationship	Description
		ReportConfigSearch.a1a2UlSearchThreshold and the ReportConfigSearch.a2UlCriticalThreshold attributes. The value of these attributes can be lowered to ensure that uplink-triggered handover takes place at lower quality. This affects all UEs in the cell. Uplink evaluation of the target cell is not performed if MBMS Multi-Carrier Support is enabled and attempts are made to move the UE to another frequency which broadcasts MBMS. If the uplink quality of the MBMS cell is poor, the Uplink-Triggered Inter-Frequency Mobility feature might attempt to move the UE back to the source cell. This can cause oscillating handover. To avoid uplink-triggered handovers, the uplink quality thresholds in the MBMS cells can be tuned down. This affects all UEs in the cell.
MBMS Multi-Carrier Support (FAJ 121 3062)	Related	The MBMS Multi-Carrier Support feature enables the UE to move to or stay on a carrier that is broadcasting an MBMS service in which the user is interested. The Uplink-Triggered Inter-Frequency Mobility does not consider whether a UE sends MBMS Interest Indication. This is to ensure that legacy behavior is followed when downlink link quality has higher priority than the ability to receive MBMS. The Uplink-Triggered Inter-Frequency Mobility feature might move the UE to another frequency with better uplink coverage, even if there is no MBMS in the cell. This occurs if the following conditions are met: – Uplink coverage in the cell is poor. – The UE is in connected mode. – The UE receives MBMS. Note: This occurs even if downlink coverage in the cell is good. UEs can combine downlink only eMBMS transmissions from multiple cells. The UE can be moved even if the user does not experience bad video quality. Note: The uplink quality thresholds are defined in the ReportConfigSearch.a1a2UlSearchThreshold and the ReportConfigSearch.a2UlCriticalThreshold attributes. The value of these attributes can be lowered to ensure that uplink-triggered handover takes place at lower quality. This affects all UEs in the cell. Uplink evaluation of the target cell is not performed if MBMS Multi-Carrier Support is enabled and attempts are made to move the UE to another frequency which broadcasts MBMS. If the uplink quality of the MBMS cell is poor, the Uplink-Triggered Inter-Frequency Mobility feature might attempt to move the UE back to the source cell. This can cause oscillating handover. To avoid uplink-triggered handovers, the uplink quality thresholds in the MBMS cells can be tuned down. This affects all UEs in the cell.

Feature	Relationship	Description
Multi-Layer Service-Triggered Mobility (FAJ 121 4124)	Related	When the Multi-Layer Service-Triggered Mobility feature is used together with Uplink-Triggered Inter-Frequency Mobility, the effect of an existing base parameter used for configuring A1 UL Search, A2 UL Search, A5-T1, A5-T2, or B2-T2 measurement threshold in the UE is changed by the corresponding offset parameter introduced in Multi-Layer Service-Triggered Mobility.
Service or Priority-Triggered Inter-Frequency Handover (FAJ 121 3087)	Related	Handover of a UE using a service that is not allowed on the source frequency cannot be performed to a target frequency on which the service is allowed. Using the Service or Priority-Triggered Inter-Frequency Handover feature with the Service or Priority-Triggered Inter-Frequency Handover feature might cause oscillating handover scenarios.
Service Triggered Mobility (FAJ 121 1747)	Related	The Service Triggered Mobility feature enables coverage-triggered mobility based on QoS. If the Service Triggered Mobility feature is used together with Uplink-Triggered Inter-Frequency Mobility, but Multi-Layer Service-Triggered Mobility is not active, and QCI based threshold offset is configured for the B2 Events, this value is applied. Values for B2-T1 threshold are not applied, since in this case the downlink quality of the serving cell is not relevant.
SRVCC Handover to GERAN (FAJ 121 3014)	Related	The SRVCC Handover to GERAN feature enables the handover of UEs with a voice bearer, that is, with a QCI value of 1, to GERAN. With the Uplink-Triggered Inter-Frequency Mobility feature enabled, this can occur when the UE reports poor uplink coverage in the current cell, or when critical uplink coverage is reported for the UE in the node.
SRVCC Handover to UTRAN (FAJ 121 2027)	Related	The SRVCC Handover to UTRAN feature enables the handover of UEs with a voice bearer, that is, with a QCI value of 1, to a UTRAN cell. With the Uplink-Triggered Inter-Frequency Mobility feature enabled, this can occur when the UE reports poor uplink coverage in the current cell, or when critical uplink coverage is reported for the UE in the node.
Mission-Critical Push-to-Talk (FAJ 121 4929)	Related	When the Mission-Critical Push-to-Talk feature is active, this feature treats mission-critical bearers like it treats VoLTE bearers (QCI 1). However, if there are restrictions on Mission-Critical Push-to-Talk service in the GERAN and UTRAN networks, it can prevent mobility towards these networks. For Release with redirect towards GERAN and UTRAN, the same handling is applied to mission-critical bearers than for VoLTE bearers, using the `UeMeasControl.allowReleaseQci1` attribute.
Uplink-Traffic-Triggered Mobility (FAJ 121 5044)	Related	When the Uplink-Triggered Inter-Frequency Mobility feature and the Uplink-Traffic-Triggered Mobility feature are enabled at the same time, the Uplink-Triggered Inter-Frequency Mobility feature uses a different criteria during UE selection for UEs after an uplink-heavy handover. It uses downlink RSRP (transmitted by the source cell during uplink-heavy handover), uplink noise and interference, and maximum UE power to calculate uplink SINR, and compares it to the uplink search threshold. The handover is only triggered if this is

Feature	Relationship	Description
		larger than the threshold.

Hardware

No specific hardware requirements.

Limitations

No known limitations.

Network Requirements

To configure redirection of the UE at poor uplink coverage, information about other carrier frequencies must be entered in the eNodeB during network configuration according to the Mobility Control at Poor Coverage feature.

3 Feature Operation of Uplink-Triggered Inter-Frequency Mobility

> This feature introduces an uplink search zone along the edge of the cell where UE measurements search for alternative frequencies and cells.

When the UE detects a frequency or cell, a mobility action is initiated. The triggering quantity for transition between the coverage zones is the uplink SINR based on DMRS power, power headroom, and noise and interference transmitted by the UE.

Note: When the Mission-Critical Push-to-Talk feature is activated, and mobility towards GERAN and UTRAN is supported, all processes and criteria in this feature operation regarding QCI 1 also apply to mission-critical bearers (QCIs 65, 66 and 69).

Feature Operation Sequence Diagram

This feature enables the eNodeB to determine where to transfer a UE that is in poor coverage. This process is enhanced by the Uplink-Triggered Inter-Frequency Mobility feature.

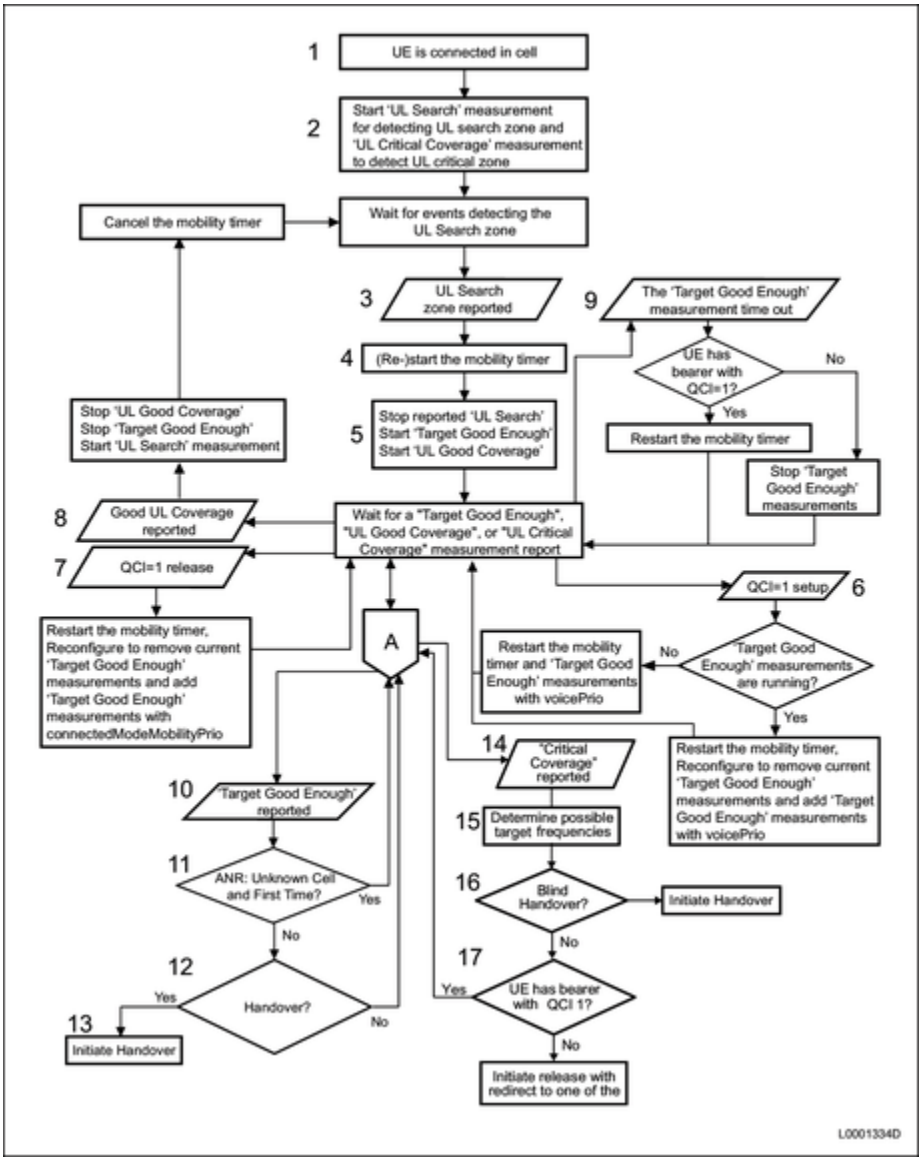

Figure 1 UE Transfer at Poor Coverage Using an Uplink Search Zone

Note: The description of the `voicePrio` and the `connectedModeMobilityPrio` attributes refer to the MO attributes on frequency relation MO classes. However, the Subscriber Triggered Mobility feature provides a way to override these values for certain UEs. For more information, see Subscriber Triggered Mobility.

Process Steps

1. The UE connects to the cell.
2. The eNodeB starts monitoring uplink coverage to detect when the serving cell drops below the threshold for entering the uplink search zone or uplink critical zone.
3. The Event A2 UL Search is triggered in the eNodeB indicating that the UE enters the uplink search zone for this triggering quantity.

   Note: ANR-triggered PCI measurements for LTE, GERAN, or UTRAN frequencies are configured when the UE attaches to the cell. These measurements are not affected by the UE entering the uplink search zone.
4. If the `UeMeasControl.a5B2MobilityTimer` attribute is set to a value different than the default `0`, the timer is started or restarted.
5. The eNodeB stops the Event A2 UL Search measurement that triggers when the UE enters the uplink search zone. The eNodeB starts the measurements that must be active in the uplink search zone. The following measurements must be active:
   - The Event A1 UL Good Coverage measurement is used to detect if the UE moves into a good uplink coverage zone.
   - The Target Good Enough measurements are used to detect the candidate frequencies. For LTE frequencies, Event A5 is used. For GERAN and UTRAN frequencies, Event B2 is used. A frequency can be measured if both the UE and the node support measurements on the frequency.

     Note: The value of the following attributes must be set to `true`:

     - `UeMeasControl.ueMeasurementsActiveIF` for LTE frequencies
     - `UeMeasControl.ueMeasurementsActiveGERAN` for GERAN frequencies
     - `UeMeasControl.ueMeasurementsActiveUTRAN` for UTRAN frequencies

     A frequency can be measured if both the UE and the node support measurements on the frequency.

     - The UE supports measurements on the frequency.
     - The node supports measurements on the frequency.

       Note: The value of the following attributes must be set to `true`:

       - `UeMeasControl.ueMeasurementsActiveIF` for LTE frequencies
       - `UeMeasControl.ueMeasurementsActiveGERAN` for GERAN frequencies
       - `UeMeasControl.ueMeasurementsActiveUTRAN` for UTRAN frequencies
     - The priority of the frequency states can be measured. A priority value of -1 means that the frequency is not measured. If the UE has an active voice bearer with a QCI value of 1, the attribute `voicePrio` is used to determine the priority. In this case, frequency relation MOs that do not have this attribute are not measured. If the UE does not have an active voice bearer with a QCI value of 1, the `connectedModeMobilityPrio` attribute is used.

       If more frequencies can be measured than those that are configured in the UE, some of the frequencies with the lowest priority are omitted according to the `voicePrio` or the `connectedModeMobilityPrio` priority attributes. This depends on whether the UE has an active voice bearer with a QCI value of 1.

     Event A5 measurements can be configured using attributes in the `ReportConfigA5UlTrig` MO.

Event B2 measurements can be configured using parameters in the `ReportConfigB2GeranUlTrig` and the `ReportConfigB2UtraUlTrig` MOs.

It is recommended to set the threshold of the `a5Threshold1Rsrp` attribute to the maximum value of -44. Using this value ensures that uplink-triggered inter-frequency handovers can occur throughout the entire cell area. In the case of B2 measurement, `b2Threshold1Rsrp` is set to the maximum value by internal parameters.

If the Automated Neighbor Relations feature is active, it uses the Target Good Enough Event A5 and Event B2 measurement reports to detect unknown cells. For more information, see *Step 12*.

6. If a bearer with a QCI value of 1 is set up while the UE is in the uplink search zone:
   - The `UeMeasControl.a5B2MobilityTimer` is restarted.
   - All Target Good Enough, Event A5 and Event B2 measurements are removed from the UE and replaced with new measurements for frequencies based on the `voicePrio` attribute. These measurements are configured in the same way as for a UE that enters the uplink search zone with a bearer with a QCI value of 1. For more information, see *Step 5*.
7. If a bearer with a QCI value of 1 is released while the UE is in the uplink search zone:
   - All Target Good Enough measurements are removed from the UE and replaced by measurements for frequencies based on the `connectedModeMobilityPrio` attribute. These measurements are configured as if the UE entered the uplink search zone without any bearer with a QCI value of 1 already set up. For more information, see step 5.
   - The `UeMeasControl.a5B2MobilityTimer` is restarted if the frequencies that are defined in the `voicePrio` attribute differ from the frequencies that are defined in the `connectedModeMobilityPrio` attribute.
8. When a UE leaves the uplink search zone and enters good uplink coverage, an Event A1 UL Good Coverage is reported in the eNodeB. The eNodeB then monitors A2 Uplink Search and A2 Uplink Critical Coverage instead of A1 UL Good Coverage. The A5 Inter-Frequency and Event B2 IRAT are removed from the UE. The `UeMeasControl.a5B2MobilityTimer` is canceled and the eNodeB waits for Event A2 UL Search or Event A2 Uplink Critical Coverage. Then, Automated Neighbor Relations can configure PCI measurements in the UE again.
9. The `UeMeasControl.a5B2MobilityTimer` expires. If the UE does not have an active bearer with a QCI value of 1, which indicates a voice bearer, the eNodeB stops all Event A5 and Event B2 Target Good Enough measurements. If the UE has an active bearer with a QCI value of 1, the eNodeB restarts the timer, but does not stop the measurements.
10. A Target Good Enough measurement report arrives from the UE for one of the frequencies.
11. The eNodeB informs the Automated Neighbor Relations feature about the best cell of the measurement report if the cell is unknown in the source cell. This is applicable even if there are other known cells in the same measurement report. This enables the Automated Neighbor Relations feature to request the UE to report the unique CGI of the potential neighbor cell. This report is sent before the next Event A5 or Event B2 Target Good Enough measurement report arrives from the UE.

    This occurs if the following requirements are fulfilled:
    - The Automated Neighbor Relations feature is activated for the specific RAT of the measurement report.
    - The best cell of the measurement report is unknown in the source cell, that is, there is no cell relation to the reported cell from the source cell.
    - No voice call is ongoing in the UE, that is, the UE has no bearer with a QCI value of 1.
    - The UE is able to do CGI measurements for the reported cell.
    - No earlier measurement report Event A5 Inter-Frequency or Event B2 IRAT Target Good Enough with best cell unknown is received. That is, for a UE, the maximum of one measurement report is ignored to allow for an Automated Neighbor Relations CGI measurement while the UE is in the search zone.
12. The eNodeB checks if a handover is allowed and possible to one of the reported cells, starting with the strongest one.

A cell is not a viable handover target in any of the following cases:

- The cell is not defined in the MOM.
- There is no cell relation to the cell from the serving cell in the MOM.
- The value of the `isHoAllowed` attribute is set to `false` on the cell relation MO.
- The uplink quality of the target cell cannot be evaluated and this is not overridden by the `ExternalENodeBFunction.ulTrigHoSupport` attribute. This applies only when an Event A5 is reported for the target cell.

  Note: If an Event B2 was reported for the target cell, the IRAT handover is performed without uplink evaluation.

- The cell has a tracking area or location area which is forbidden according to the HRL of the UE.

If the `UeMeasControl.bothA5RsrpRsrqCheck` attribute is set to `true`, the eNodeB uses an extended check of A5 Target Good Enough measurements before handover based on the following conditions:

- RSRP measured value must be higher than either of the following calculated expressions:
  - $a5Threshold2Rsrp + \max_{ieUE_QCISet}[a5Threshold2RsrpOffset(qciPr\ ofile_i)] + hysteresisA5 - cellIndivOffset_C - qOffsetFreq_F$

    The term $\max_{ieUE_QCISet}[a5Threshold2RsrpOffset(qciPr\ ofile_i)]$ is included only if the feature Service Triggered Mobility is operable and the feature Multi-Layer Service-Triggered Mobility is not activated. The `cellIndivOffset_C` attribute applies to Data CIO or QCI1 CIO, depending on the configured QCI1 bearer and whether the `EUtranFreqRelation.cioQci1Enabled` attribute is set to `true`.
  - $a5Threshold2Rsrp + a5Thr2RsrpFreqOffset + \max_{ieUE_QCISet}[a5Thr2RsrpFreqQciOffset(qciProfilei)] + hysteresisA5 - cellIndivOffset_C - qOffsetFreq_F$

    The sum $a5Thr2RsrpFreqOffset + \max_{ieUE_QCISet}[a5Thr2RsrpFreqQciOffset(qciProfile_i)]$ is included only if the feature Multi-Layer Service-Triggered Mobility is operable. The `cellIndivOffset_C` attribute applies to Data CIO or QCI1 CIO, depending on the configured QCI1 bearer and whether `EUtranFreqRelation.cioQci1Enabled` is activated.
- RSRQ measured value must be higher than the following calculated expression:

  $a5Threshold2Rsrq + \max_{ieUE_QCISet}[a5Threshold2RsrqOffset(qciPr\ ofile_i)] + hysteresisA5 - cellIndivOffset_C - qOffsetFreq_F$

  The `cellIndivOffset_C` parameter applies to Data CIO or QCI1 CIO, depending on the configured QCI1 bearer and whether the `EUtranFreqRelation.cioQci1Enabled` attribute is set to `true`.

13. If a cell is selected, the handover is initiated.

    In case of IRAT handover, uplink evaluation is not performed.

    Note: If the selected cell is a WCDMA cell, oscillating IRAT handover can occur. To avoid this, uplink quality thresholds must be tuned down or B2 thresholds must be tuned up. This affects all UEs in the cell.

    If the target cell supports the Uplink-Triggered Inter-Frequency Mobility feature, uplink channel evaluation in the target cell is performed for X2 handovers. The evaluation takes place when the target cell receives the handover request message. The following requirements must be fulfilled for the handover to be performed:

    - The UE must not arrive to UL search state in the target cell:

      $SINRmax_{RSRP}(target) \geq targetA2UlSearchOffset + a1a2UlSearchThreshold(target)$

      The `UeMeasControl.targetA2UlSearchOffset` attribute is configured on the target side.
    - The UL quality must be an offset better in the target than in the source cell:

      $SINRmax_{RSRP}(target) \geq SINRmax_{RSRP}(source) + ulSinrOffset$

      The `UeMeasControl.ulSinrOffset` attribute is configured on the source side.

SINRmaxRSRP is based on UE reported RSRP and measured noise+interference (N+I) on the uplink carrier.

Note: If the High Power UE feature is used, the time normalized filtered SINR calculation for DMRS and the SINR calculation in uplink evaluation that is based on the RSRP of the source cell consider the following:

- Carrier Aggregation configuration
- UE power class

If the handover is accepted, the `Handover Request Acknowledge` X2 message is sent to the source eNodeB.

If the handover is rejected, the `Handover Preparation Failure` X2 message is sent to the source eNodeB with applicable cause value.

– If the UE arrives to uplink search state in the target cell, the target eNodeB rejects the incoming handover with the `targetUlBelowThreshold` cause.

– If the uplink quality is not an offset better in the target than in the source cell, the target eNodeB rejects the incoming handover with the `ulImprovementBelowThreshold` cause.

In case none of the reported cells are viable for handover, the eNodeB takes no further action. This means that the UE remains in the cell and waits for another Target Good Enough measurement report or an Event A1 Good Uplink Coverage.

For more information, see Manage Radio Network and Manage Mobility.

14. An Event A2 UL Critical Coverage is reported in the node.

15. A prioritized list of possible target frequencies is determined for handover if the UE reaches critical coverage.

    a. The node determines a set of LTE inter-frequency or IRAT frequencies where the UE can be transferred when it encounters poor coverage in the current cell. The node finds all applicable frequency relation MO instances for the current cell:

    – `EUtranFreqRelation`

    – `GeranFreqGroupRelation`

    – `UtranFreqRelation`

    – `UtranTDDFreqRelation`

    – `Cdma2000FreqBandRelation`

    A frequency relation MO instance is applicable if the following conditions are met:

    – The corresponding Coverage-Triggered Session Continuity feature is active and enabled.

    Table 2 Coverage-Triggered Session Continuity Features for Redirection

Target RAT	Feature
LTE	Coverage-Triggered Inter-Frequency Session Continuity (FAJ 121 0797)
WCDMA	Coverage-Triggered WCDMA Session Continuity (FAJ 121 0493)
GERAN	Coverage-Triggered GERAN Session Continuity (FAJ 121 0495)
CDMA2000	Coverage-Triggered CDMA-eHRPD Session Continuity (FAJ 121 0491)

    – The UE is able to operate on the frequency band according to UE capabilities. This is reported to the node in the UE-EUTRA-Capability IE.

    Note: When the Multiple Frequency Band Indicators feature is operable, the UE can support more

frequencies.

- The corresponding RAT is not forbidden according to the HRL of the UE.
- If the `allowedPlmnList` attribute of the frequency relation MO is not empty, that is, at least one of the PLMNs in this attribute matches one of the PLMNs in the HRL of the UE.

b. If the `UeMeasControl.excludeInterFreqAtCritical` attribute is set to `true`, all E-UTRAN frequencies are removed from the list.

c. The used priority is decided.

Table 3 Frequency Relation Priority Selection

QCI of Active UE Bearer	Used Attribute
The UE has an active bearer with QCI=1	`voicePrio` Any frequency relation that does not have a `voicePrio` attribute is considered to have priority −1.
The UE has no active bearer with QCI=1	`connectedModeMobilityPrio`

d. The list is ordered according to the selected priority, and each frequency with priority −1 is removed from the list.

e. If a frequency has been measured in the search zone for longer than the time indicated by the value of the `UeMeasControl.searchEffortTime` attribute, it is excluded from the list as non-feasible target. If this would leave the list empty, the list is left unmodified.

If the value of the `UeMeasControl.searchEffortTime` attribute is `0`, the list is always unmodified by this step.

16. The eNodeB checks if a blind handover is possible to a cell on one the high priority frequencies in the list of possible target frequencies.

A cell is a viable blind handover target if the following requirements are fulfilled:

- The `mobilityAction` attribute on the relevant frequency relation MO has the value `HANDOVER`.
- The relevant handover feature is operable.

Table 4 Handover Features for Blind Handover

Target RAT	Feature
LTE	Coverage-Triggered Inter-Frequency Handover (FAJ 121 0877)
	Intra-LTE Inter-Mode Handover (FAJ 121 3042) Note: This feature is applicable when the frequency has a different mode, TDD or FDD.
WCDMA	Coverage-Triggered WCDMA IRAT Handover
	SRVCC Handover to UTRAN
GERAN	SRVCC Handover to GERAN

- The UE has support for handover to the relevant RAT.
- The cell is defined in the MOM.
- There is a cell relation to the target cell from the serving cell in the MOM.
- The `isHoAllowed` attribute is set to `true` on the cell relation MO.
- The `coverageIndicator` attribute is set to `COVERS` on the cell relation MO.
- The `EUtranCellFDD.covTriggerdBlindHoAllowed` or the

EUtranCellFDD.covTriggerdBlindHoAllowed attribute is set to true on the serving cell.

- The cell has a tracking area or location area that is not forbidden according to the HRL of the UE.
- The feature Cell Soft Lock has not temporarily set the cell as a non-viable target.

If a cell is selected, the handover is initiated without uplink evaluation.

17. If the UE has an active bearer with a QCI value of 1, which indicates a voice bearer, UE release is not allowed. The node takes no further action and the UE remains in the cell and waits for one of the following measurement reports:
    - Target Good Enough
    - Good Coverage
    - Critical Coverage
    - Search Zone

    This behavior can be changed by setting the allowReleaseQci1 attribute to true.

    Note: Setting the UeMeasControl.allowReleaseQci1 attribute takes effect only if SRVCC Handover to UTRAN or SRVCC Handover to GERAN is not operable.

    If the eNodeB performs a release with redirect, the default behavior in a Core Network is to remove the voice connection.

    If the UE does not have an active bearer with QCI=1, the node initiates a release with redirect to the frequency with the highest priority in the list of possible target frequencies.

3.1 Measurement Configuration

> The uplink search zone follows the edge of the cell and is detected by Event A2 UL measurements.

Figure 2 shows the uplink search zone.

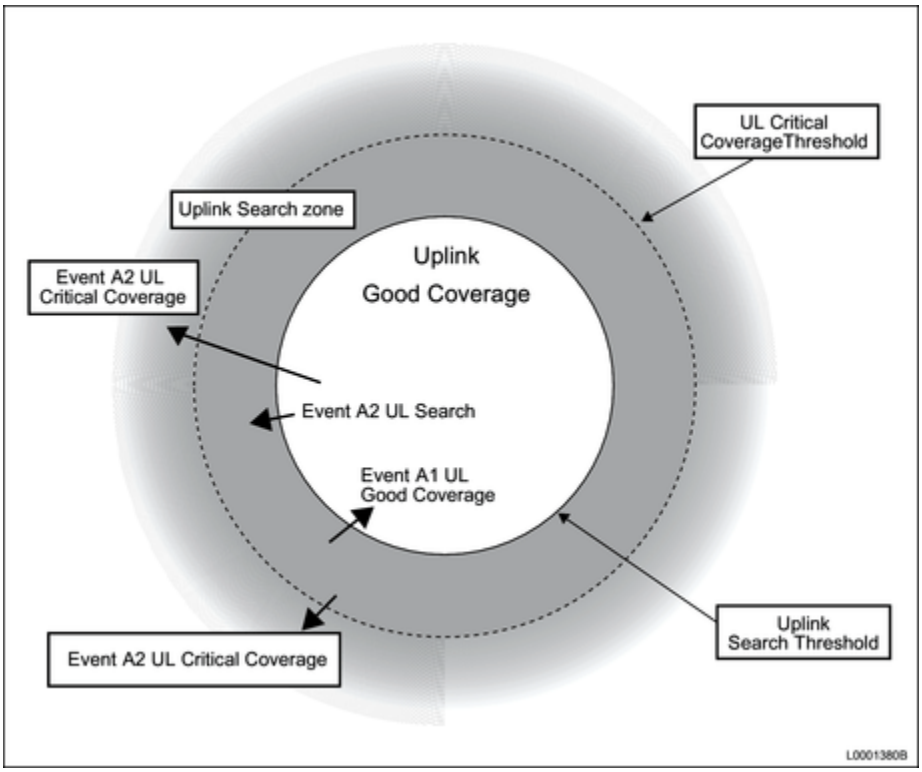

Figure 2 Cell Uplink Search Zone Defined by Uplink Search Thresholds

The Search Threshold

To detect that the UE has entered the uplink search zone, Event A2 UL Search measurements are started in the eNodeB when it attaches to the cell. When an Event A2 UL Search measurement triggers, the UE is considered to be in the uplink search zone. To detect when the UE leaves the uplink search zone and enters into good uplink coverage again, corresponding Event A1 UL Good Coverage measurements are started in the eNodeB.

The report configuration for both the Event A2 UL Search and Event A1 UL Good Coverage is based on parameters in the `ReportConfigSearch` MO.

The report configurations for these events are also affected by the Service Triggered Mobility and the Multi-Layer Service-Triggered Mobility features.

It is not recommended to set the search threshold of the `a1a2UlSearchThreshold` attribute higher than `320`. This causes UEs to trigger Event A2 UL and move to the uplink search zone after attaching to the cell. Once the UEs are in the uplink search zone, the UEs do not trigger Event A1 UL and do not return to the uplink good coverage zone. Configurations of the `ReportConfigSearch.qciA1A2UlThrOffsets` attribute that decrease the effective value of the search threshold under `320` can prevent this behavior for configured QCIs.

The Critical Threshold

To detect that the UE has encountered uplink critical coverage, Event A2 UL Critical Coverage measurements are started on the node when the UE attaches to the cell. The report configuration for this is based on parameters in the `ReportConfigSearch` MO.

The report configurations for this measurement are not affected by the Service Triggered Mobility and Multi-Layer Service-Triggered Mobility features.

4 Network Impact of Uplink-Triggered Inter-Frequency Mobility

> This feature enhances network capacity by moving UEs to the best available frequency in the network.

Capacity and Performance

Moving UEs to different frequencies improves coverage, however, the feature has no impact on physical cell coverage. When the feature is enabled, UEs can be moved to cells in different frequencies with better radio conditions more often.

When the feature is enabled together with the High Power UE feature, uplink coverage is improved by moving high power UEs to cells where high power UE capability is allowed.

The number of handovers to LTE and IRAT cells is increased at poor uplink radio conditions for LTE cells. Uplink radio conditions can be measured independently of downlink radio conditions. This means that that unnecessary handovers are avoided.

For a UE with critical uplink radio conditions in the cell, a blind handover is attempted to an LTE or an IRAT cell. If handover is not possible, the UE is released and redirected to a better LTE or IRAT cell. The number of abnormal releases because of poor uplink radio conditions is decreased.

Interfaces

No impact.

Other Network Elements

No impact.

5 Parameters for Uplink-Triggered Inter-Frequency Mobility

> This feature introduces parameters related to detecting the Uplink Search Threshold.

Table 5 Parameters

Parameter	Type	Description
ReportConfigSearch.a1a2UlSearchThreshold	Introduced	See MOM description.
ReportConfigSearch.hysteresisA1A2UlSearch	Introduced	See MOM description.
ReportConfigSearch.timeToTriggerA1UlSearch	Introduced	See MOM description.
ReportConfigSearch.timeToTriggerA2UlSearch	Introduced	See MOM description.
ReportConfigSearch.qciA1A2UlThrOffsets	Introduced	See MOM description. This struct contains the following members: – a1a2UlThrQciOffset – qciProfileRef
ReportConfigSearch.a2UlCriticalThreshold	Introduced	See MOM description.
ReportConfigSearch.hysteresisA2UlCritical	Introduced	See MOM description.
ReportConfigSearch.timeToTriggerA2UlCritical	Introduced	See MOM description.
ExternalENodeBFunction.ulTrigSupport	Introduced	See MOM description.
EUtranCellFDD.ulTrigActive	Introduced	See MOM description.
EUtranCellTDD.ulTrigActive	Introduced	See MOM description.
ENodeBFunction.enabledUlTrigMeas	Introduced	See MOM description.
UeMeasControl.targetA2UlSearchOffset	Introduced	See MOM description.
UeMeasControl.ulSinrOffset	Introduced	See MOM description.
EUtranFreqRelation.interFreqMeasTypeUlSinr	Introduced	See MOM description.
ExternalENodeBFunction.ulTrigHoSupport	Introduced	See MOM description.
ReportConfigA5UlTrig.a5Threshold1Rsrp	Introduced	See MOM description. It is recommended to set this attribute to the highest possible value, which equals the default value, to ensure that the downlink source cell signal strength does not affect the Uplink-Triggered Inter-Frequency Mobility.
ReportConfigA5UlTrig.a5Threshold2Rsrp	Introduced	See MOM description. It is recommended to have this threshold set to the same value as ReportConfigA5.a5Threshold2Rsrp.
ReportConfigA5UlTrig.hysteresisA5	Introduced	See MOM description.
ReportConfigA5UlTrig.reportIntervalA5	Introduced	See MOM description.
ReportConfigA5UlTrig.timeToTriggerA5	Introduced	See MOM description.
ReportConfigB2GeranUlTrig.b2Threshold2Geran	Introduced	See MOM description.
ReportConfigB2GeranUlTrig.hysteresisB2	Introduced	See MOM description.
ReportConfigB2GeranUlTrig.reportIntervalB2	Introduced	See MOM description.
ReportConfigB2GeranUlTrig.timeToTriggerB2	Introduced	See MOM description.
ReportConfigB2UtraUlTrig.b2Threshold2EcNoUtra	Introduced	See MOM description.
ReportConfigB2UtraUlTrig.b2Threshold2RscpUtra	Introduced	See MOM description.
ReportConfigB2UtraUlTrig.hysteresisB2	Introduced	See MOM description.

Parameter	Type	Description
ReportConfigB2UtraUlTrig.reportIntervalB2	Introduced	See MOM description.
ReportConfigB2UtraUlTrig.timeToTriggerB2	Introduced	See MOM description.
UeMeasControl.a5B2MobilityTimer	Affecting	See MOM description.
UeMeasControl.searchEffortTime	Affecting	See MOM description.

6 Performance of Uplink-Triggered Inter-Frequency Mobility

> This feature improves integrity KPIs.

KPIs

The following KPIs are improved:

- Packet Loss Rate
- UL Packet Loss Rate per QCI
- VoIP Integrity
- UL Throughput
- DL Throughput

Counters

The following counters are associated with this feature:

-
  - `EUtranCellFDD.pmGoodCovSearchEvalReportUl`
  - `EUtranCellTDD.pmGoodCovSearchEvalReportUl`
  - `EUtranCellFDD.pmBadCovSearchEvalReportUl`
  - `EUtranCellTDD.pmBadCovSearchEvalReportUl`
  - `EUtranCellFDD.pmCriticalBorderEvalReportUl`
  - `EUtranCellTDD.pmCriticalBorderEvalReportUl`
  - `EUtranCellFDD.pmCovGoodZoneSum`
  - `EUtranCellTDD.pmCovGoodZoneSum`
  - `EUtranCellFDD.pmCovGoodZoneSamp`
  - `EUtranCellTDD.pmCovGoodZoneSamp`
  - `EUtranCellFDD.pmCovSearchZoneSum`
  - `EUtranCellTDD.pmCovSearchZoneSum`
  - `EUtranCellFDD.pmCovSearchZoneSamp`
  - `EUtranCellTDD.pmCovSearchZoneSamp`
  - `EUtranCellFDD.pmA5BestCellEvalReportUl`
  - `EUtranCellTDD.pmA5BestCellEvalReportUl`
  - `EUtranCellFDD.pmB2BestCellEvalReportUl`
  - `EUtranCellTDD.pmB2BestCellEvalReportUl`
  - `EUtranCellFDD.pmCellHoPrepAttLteInterFUl`
  - `EUtranCellTDD.pmCellHoPrepAttLteInterFUl`
  - `EUtranCellFDD.pmCellHoExeAttLteInterFUl`
  - `EUtranCellTDD.pmCellHoExeAttLteInterFUl`
  - `EUtranCellFDD.pmCellHoExeSuccLteInterFUl`

- EUtranCellTDD.pmCellHoExeSuccLteInterFUl
- EUtranCellFDD.pmHoPrepRejOutUlThres
- EUtranCellTDD.pmHoPrepRejOutUlThres
- EUtranCellFDD.pmUeCtxtRelSCEutraUlSinr
- EUtranCellTDD.pmUeCtxtRelSCEutraUlSinr
- EUtranCellFDD.pmUeCtxtRelSCUtraUlSinr
- EUtranCellTDD.pmUeCtxtRelSCUtraUlSinr
- EUtranCellFDD.pmUeCtxtRelSCGsmUlSinr
- EUtranCellTDD.pmUeCtxtRelSCGsmUlSinr
- EUtranCellFDD.pmUeCtxtRelSCCdmaUlSinr
- EUtranCellTDD.pmUeCtxtRelSCCdmaUlSinr
- EUtranCellRelation.pmHoPrepAttLteInterFUl
- EUtranCellRelation.pmHoPrepSuccLteInterFUl

Events

Event	Event Parameter
INTERNAL_EVENT_UE_MOBILITY_EVAL	EVENT_PARAM_SINR_MAX_DMRS EVENT_PARAM_THRESHOLD_RSRP EVENT_PARAM_THRESHOLD_RSRQ EVENT_PARAM_THRESHOLD_UL_SINR_MAX EVENT_PARAM_EST_POWER_HEADROOM EVENT_PARAM_DL_COV_STATE EVENT_PARAM_UL_COV_STATE EVENT_PARAM_FILTERED_EQ_GAIN EVENT_PARAM_ULCOMP_COMBINING_SECTOR_CELL_ID EVENT_PARAM_TTI_BUNDLING_MODE
INTERNAL_PROC_HO_PREP_X2_OUT	EVENT_PARAM_UL_EVALUATION EVENT_PARAM_SINR_MAX_RSRP_SOURCE EVENT_PARAM_NOISE_INTERFERENCE_SOURCE EVENT_PARAM_SINR_MAX_RSRP_TARGET EVENT_PARAM_NOISE_INTERFERENCE_TARGET EVENT_PARAM_ULTRIG_REJECT_CAUSE

7 Activate Uplink-Triggered Inter-Frequency Mobility

> To use the feature, it must be activated following the usual feature activation procedure.

Prerequisites

- The license key is installed in the node.
- CCTR is active for at least one week before this procedure to collect enough troubleshooting data.
- The Mobility Control at Poor Coverage feature is activated.

Steps

1. Set the `FeatureState.featureState` attribute to `ACTIVATED` in the `FeatureState=CXC4011072` MO instance.

After This Task

Keep CCTR active for at least one week after this procedure for continued collection of troubleshooting data.

8 Deactivate Uplink-Triggered Inter-Frequency Mobility

> If the feature is no longer needed, it can be deactivated following the usual feature deactivation procedure. It must also be deactivated before the activation of any conflicting feature.

Prerequisites

CCTR is active for at least one week before this procedure to collect enough troubleshooting data.

Steps

1. Set the `FeatureState.featureState` attribute to `DEACTIVATED` in the `FeatureState=CXC4011072` MO instance.

After This Task

Keep CCTR active for at least one week after this procedure for continued collection of troubleshooting data.

9 Engineering Guidelines for Uplink-Triggered Inter-Frequency Mobility

> This feature must be configured to ensure that it is operable.

The following requirements must be fulfilled at feature activation:

- The Mobility Control at Poor Coverage feature must be activated.
- To configure UE redirection at poor uplink coverage, information about other carrier frequencies must be entered in the eNodeB during network configuration according to the Mobility Control at Poor Coverage feature.
- The settings must be adjusted for the following events so that they are internally reported in the eNodeB when required:

Table 6 Events Reported in the Node

Event	Description
Event A1 Uplink Good Coverage	The event detects when the UE has good uplink coverage.
Event A2 Uplink Search	The event detects when the UE enters the uplink search zone.
Event A2 Uplink Critical Coverage	The event detects when the UE has critical uplink coverage.

When the UE enters the uplink search zone, the eNodeB can configure Event A5 measurements in the UE to search for alternative frequencies. This is controlled by the `UeMeasControl.ueMeasurementsActiveIF` attribute.

When a UE enters the uplink search zone, the eNodeB can configure Event B2 measurements in the UE to search for alternative RANs. This is controlled by the `UeMeasControl.ueMeasurementsActiveGERAN` and the `UeMeasControl.ueMeasurementsActiveUTRAN` attributes.

Observability of the uplink quality in the network can be enabled without a license for the feature by configuring the `ENodeBFunction.enabledUlTrigMeas` attribute.

Appendix A: Feature Change History

> This section lists changes that affected this feature and the impact it had on the network.

Appendix A.a: 17.Q4: IRAT Handover at Bad Uplink Radio Conditions

> The enhancement of Uplink-Triggered Inter-Frequency Mobility enables better retainability and mobility in network deployment scenarios with uplink problems where IRAT cells are available.

Feature Name:	Uplink-Triggered Inter-Frequency Mobility
Feature Identity:	FAJ 121 1797 R2
Value Package Name:	LTE Base Package
Value Package Identity:	FAJ 801 0400 R9
Node Type:	Baseband Radio Node, DU Radio Node
Access Type:	LTE

The enhancement of Uplink-Triggered Inter-Frequency Mobility introduces the possibility of IRAT handover to UTRAN or GERAN, by enabling to configure a UE with B2 IRAT Event in bad uplink radio conditions. This results in better retainability and mobility in network deployment scenarios with uplink problems where IRAT cells are available.

Note: In case of uplink-triggered IRAT handover no uplink evaluation of the IRAT target cell is made.

Capacity and Performance

The number of handover to IRAT cells is increased at bad uplink radio conditions for LTE cells.

This decreases packet loss rate due to bad uplink radio conditions and per QCI, improves throughput and VoIP integrity.

The benefits of the enhancement can be observed using counter `pmB2BestCellEvalReportUl`.

Operation

New MO Classes are introduced for configuration of B2 Event thresholds:

- `ReportConfigB2UtraUlTrig` to configure UTRA B2 Event measurements
- `ReportConfigB2GeranUlTrig` to configure GERAN B2 Event measurements

The following MO Classes are removed, as handover to CDMA2000 is not supported:

- `ReportConfigB2CdmaUlTrig`
- `ReportConfigB2CdmaRttUlTrig`

Interfaces

The change affects the RRC protocol on the Uu interface.

IRAT Event B2 can be started on the UE in the UL Search Zone. This affects the following information elements for UTRA and GERAN measurements within `ReportConfigInterRAT` information element:

- `b2-Threshold1`
- `hysteresis`
- `timeToTrigger`
- `maxReportCells`
- `reportInterval`

- `reportAmount`
- `ThresholdUTRA`
  - `utra-RSCP`
  - `utra-EcN0`
- `ThresholdGERAN`

Hardware

No impact.

Other Network Elements

No impact.

Appendix A.b: 17.Q4: Mobility Actions at Critical Uplink Quality

> Uplink-Triggered Inter-Frequency Mobility enhancement introduces the possibility of blind handover and release with redirect to E-UTRAN inter-frequency or IRAT cells at a configured critical coverage threshold.

Feature Name:	Uplink-Triggered Inter-Frequency Mobility
Feature Identity:	FAJ 121 1797 R2
Value Package Name:	LTE Base Package
Value Package Identity:	FAJ 801 0400 R9
Node Type:	Baseband Radio Node, DU Radio Node
Access Type:	LTE

The enhancement of Uplink-Triggered Inter-Frequency Mobility introduces the possibility of blind handover and release with redirect to E-UTRAN inter-frequency or IRAT cells at a configured critical coverage threshold. This results in better retainability and integrity in network deployment scenarios with uplink problems.

Note: In case of blind handover or release with redirect due to critical uplink quality, no uplink evaluation of the target cell is made.

Capacity and Performance

The number of handover to LTE or IRAT cells is increased at critical uplink radio conditions for LTE cells.

This decreases packet loss rate due to bad uplink radio conditions and per QCI, improves throughput and VoIP integrity.

The number of release with redirect to LTE or IRAT cells is increased at critical uplink radio conditions for LTE cells.

The benefits of the enhancement can be observed using the following counters:

- `EUtranCellFDD.pmUeCtxtRelSCEutraUlSinr`
- `EUtranCellTDD.pmUeCtxtRelSCEutraUlSinr`
- `EUtranCellFDD.pmUeCtxtRelSCUtraUlSinr`
- `EUtranCellTDD.pmUeCtxtRelSCUtraUlSinr`
- `EUtranCellFDD.pmUeCtxtRelSCGsmUlSinr`
- `EUtranCellTDD.pmUeCtxtRelSCGsmUlSinr`
- `EUtranCellFDD.pmUeCtxtRelSCCdmaUlSinr`
- `EUtranCellTDD.pmUeCtxtRelSCCdmaUlSinr`

Operation

No impact.

Interfaces

No impact.

Hardware

All DU and Baseband units.

Other Network Elements

No impact.

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